

JAMESTOWN, ND, USA

WMO: 727535

Lat: 46.926N

Lon: 98.669W

Elev: 1494

StdP: 13.92

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14919

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-13.5	-23.7	1.6	-17.4	-19.1	2.0	-13.1	33.3	15.4	30.6	13.9	9.4	310	0.690

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	89.2	72.2	86.0	69.9	83.1	68.1	75.1	84.7	72.9	82.3	70.7	79.9	12.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	126.0	80.8	69.8	115.9	79.0	67.3	106.3	76.3	39.6	84.8	37.5	82.4	35.5	79.7	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.9	25.9	23.2	DB	-24.0	95.7	5.5	3.4	-28.0	98.1	-31.2	100.1	-34.3	102.0	-38.3	104.4	(4)
(5)			WB	-24.3	78.9	5.4	2.4	-28.1	80.6	-31.3	82.0	-34.3	83.4	-38.2	85.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.7	10.1	13.6	26.8	41.7	54.7	65.3	70.2	67.7	59.1	44.5	29.0	15.8		
	DBStd	23.54	14.26	13.65	13.28	10.52	8.64	6.78	5.41	5.77	8.39	9.33	11.38	12.73		
	HDD50	5235	1236	1018	721	281	51	1	0	0	17	220	631	1059		
	HDD65	8941	1701	1438	1183	699	333	78	16	38	215	637	1079	1524		
	CDD50	2206	0	0	3	33	197	459	625	548	289	50	2	0		
	CDD65	438	0	0	0	1	14	85	177	122	37	2	0	0		
	CDH74	4530	0	0	1	25	227	856	1698	1221	457	45	0	0		
	CDH80	1401	0	0	0	4	52	237	562	393	143	10	0	0		
(14) Wind	WSAvg	11.0	11.7	11.8	12.0	12.4	12.2	10.4	8.8	8.6	9.8	11.4	11.5	11.6		
(15) Precipitation	PrecAvg	17.8	0.6	0.5	0.9	1.5	2.2	3.3	3.0	2.1	1.7	1.0	0.5	0.4		
	PrecMax	30.2	2.2	2.2	3.5	3.1	6.2	7.6	11.1	5.7	5.3	4.3	2.4	1.2		
	PrecMin	9.6	0.0	0.0	0.0	0.0	0.2	0.9	0.4	0.3	0.0	0.0	0.0	0.0		
	PrecStd	4.2	0.5	0.6	0.8	0.9	1.6	1.7	2.2	1.4	1.3	1.0	0.6	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.2	46.0	65.8	77.6	85.8	90.7	93.5	92.9	90.1	80.5	63.5	45.7		
		MCWB	36.7	39.0	51.8	56.6	65.4	72.1	73.3	72.6	70.2	61.6	50.5	38.7		
	2%	DB	38.2	40.1	56.6	70.3	80.5	85.8	89.1	88.1	83.9	71.6	55.2	40.0		
		MCWB	34.2	35.9	47.1	53.0	61.9	69.0	73.5	70.9	66.1	57.2	45.4	35.1		
	5%	DB	34.7	36.4	49.3	64.6	75.3	82.3	85.9	84.2	79.3	64.7	49.6	36.2		
		MCWB	31.9	33.6	43.4	50.0	59.2	66.5	71.4	69.5	64.0	53.2	42.5	32.7		
	10%	DB	30.8	33.6	43.4	59.4	71.0	79.2	82.7	81.1	74.6	59.5	44.2	32.5		
		MCWB	29.0	31.7	39.0	47.4	56.9	64.7	69.6	67.6	62.0	51.0	38.6	30.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.2	39.9	54.2	58.4	68.6	76.2	78.8	77.1	72.8	63.4	51.4	39.2		
		MCDB	41.6	44.7	61.6	73.1	80.5	85.8	87.7	86.5	85.3	75.3	61.3	44.9		
	2%	WB	34.5	36.5	48.0	54.1	64.8	72.5	75.8	74.0	69.4	58.7	46.7	35.4		
		MCDB	37.7	39.9	56.2	66.9	75.7	81.9	85.0	84.0	79.9	68.6	54.3	39.3		
	5%	WB	32.2	33.6	43.1	51.2	61.6	69.4	73.6	71.4	66.7	54.7	42.6	32.9		
		MCDB	34.7	35.9	49.2	63.1	71.1	79.2	82.5	80.6	75.6	63.7	49.3	35.7		
	10%	WB	28.9	31.2	39.0	48.5	58.9	67.0	71.4	69.3	64.0	51.0	38.7	30.3		
		MCDB	30.8	33.7	43.4	58.4	68.1	75.8	80.0	78.3	72.2	58.6	43.7	32.5		
(35) Mean Daily Temperature Range	MDBR		16.5	17.2	17.7	21.2	22.0	20.4	21.6	22.8	23.1	19.7	17.4	15.3		
	5% DB	MCDBR	19.7	19.0	26.7	31.6	28.6	25.6	25.8	27.5	30.4	30.3	26.3	20.5		
		MCWBR	16.9	15.1	18.1	16.9	14.4	12.1	12.6	13.3	14.8	17.4	17.3	16.4		
	5% WB	MCDBR	19.2	17.9	26.0	28.9	23.6	22.5	22.7	24.3	25.6	27.6	25.3	19.5		
		MCWBR	16.7	14.6	18.1	16.1	13.4	12.5	13.0	13.6	14.4	16.8	17.2	16.0		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.313	0.295	0.364	0.376	0.377	0.379	0.383	0.350	0.302	0.270	0.255		
	taud		2.248	2.254	2.461	2.287	2.331	2.377	2.380	2.355	2.425	2.537	2.509	2.446		
	Ebn at Noon		245	269	297	282	282	281	279	273	273	275	265	258		
	Edh at Noon		25	31	30	39	39	37	36	36	31	23	19	18		
(44) All-Sky Solar Radiation	RadAvg		533	889	1246	1570	1795	1990	2068	1736	1311	823	517	392		
	RadStd		47	92	102	123	132	135	101	134	86	103	68	35		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page